

A data driven recurrent neural network approach for reproduction of variable visuo-haptic force feedback in surgical tool insertion

Introduction

NEEDLE or surgical tool insertion procedures play a significant role in Robotic-Assisted Minimally Invasive Surgery (RAMIS). It is a skill imperative for surgeons for the adequate performance of any surgical task. There are four major situations where the clinician is likely to make mistakes during the insertion task: the determination of the exact tool insertion point, the orientation of the tool, the movement of the tool into soft tissues, and the probability of a physician getting diverted from the target due to motion artifacts, mental stress, and unconditional tissue deformation. Other inaccuracies happen due to the variable tissue parameters in different patients, which change from person to person, age group to age group, location to location, live and dead cells, and changes in the physiology of the organ in planning and treatment stages. These constraints increase the likelihood of needles or surgical tools passing through sensitive tissues such as arteries, bones, nerves, and organs, potentially causing significant tissue damage and unpredictable medical accidents resulting in intense pain (Mahmood et al., 2022, Ravali and Manivannan, 2017, Spinoglio et al., 2015). Therefore, surgeons require constant practice for avoiding such incidents and ensure accuracy in the surgical procedure. This work focuses on two different categories of people: the surgeon, who can practice in a real-time surgical environment, and the medical student or trainee, who can practice surgical procedures using Visuo-haptic feedback in a simulated environment.

In the classical method, surgeons use their fingers to palpate tissue to get an understanding of the properties of the tissue (Coles et al., 2011) and perform a surgical task. However, tactile sensors are not always effective for providing tactile feedback during minimally invasive surgery (MIS) because they do not make direct contact with soft tissue (Gwilliam et al., 2010). Advanced surgical systems, such as the Intuitive surgical da Vinci system (Fontanelli et al., 2017, Guthart and Salisbury, 2000), offer higher dexterity, accuracy, effectiveness, and control during precise surgical tasks

(Darzi and Munz, 2004, Enayati et al., 2016, Spinoglio et al., 2015), and provide more realistic haptic feedback to the system. To determine the exact tool-tissue interaction force, researchers use a combination of tactile and force sensors (Dargahi et al., 2007; Liu et al., 2010; Sabatini et al., 1990) or sensorized surgical tools (Talasaz and Patel, 2013, Wee et al., 2017, Zarrin et al., 2018). However, none of the models have been clinically validated. The researcher (Liu et al., 2011) developed an effective method for determining stiffness using discrete palpation and implemented a control system in the da Vinci system using a position-position controller with adequate compensation for inertia and friction. Some of the works (Xu and Simaan, 2008, Xu and Simaan, 2010) demonstrate the use of joint-level actuation forces for the estimation of the tooltip forces for the evaluation of stiffness.

The researchers (Bano et al., 2013) estimated the stiffness on the basis of preoperative surface data registration and vision-based force estimation (Aviles et al., 2017, Greminger and Nelson, 2003, Haouchine et al., 2018, Schoob et al., 2016) by reconstructing the surface geometry and stereo vision-based depth mapping. The work presented by (Chalasani et al., 2018) shows the stiffness estimation and the reconstruction (Correal et al., 2014) of organ geometry performed online using force-torque sensor data. Some notable works (Ayvali et al., 2016) have explored stiffness using Bayesian optimization strategies by guiding the probe to the unknown region for maximum information gain. (Caccamo et al., 2016), A framework was set for the probabilistic approach of acquiring data from heterogeneous elastic surfaces and deformation distribution maps subjected to tool tissue interactions using haptics and visual information. Some recent works (Gao et al., 2018, Marban et al., 2019, Mendizabal et al., 2019) show force estimation and prediction based on training advanced neural networks such as LSTM-based recurrent neural network (Aviles et al., 2017), convolution neural network (Marban et al., 2019), and RGB point cloud temporal convolutional neural network (Gao et al., 2018) through the acquisition of depth, visual deformation information, and tool data, etc.

A new algorithm has been proposed by (Abeywardena et al., 2019) for estimating interaction forces using the motor current and neural network. Only 3D-printed materials of varying hardness and form were used for the validation of the method. However, the work has a significant limitation in that it has no application to biological tissue at this stage due to variable biological tissue properties in different layers. Edwards et al. (2021) proposed visual kinematic force estimation based on kinematics and visual tool position measurement discrepancy. This method has the advantage of not requiring any modeling or extrapolation. The authors demonstrated the ability to discern force in knot tying using the JIGSAWS dataset and examples from the da Vinci

surgical system. The system, however, does not have the ability to measure absolute force due to its requirement of physical modeling of the robot as well as 3D visualization of the surgical instrument. Jung et al. (2021) have proposed a vision-based suture tensile force estimation method that uses a single image and robot position without the use of an external force sensor. As a suture-pulling force prediction (Ehrampoosh et al., 2022) tool, this study used modified Inception Resnet-V2 and LSTM networks. The results were validated by using two artificial skins with different elastic moduli. However, due to the dynamic nature of the environment, the method relays only a single image with a limited feature set and fails when piercing the tool into the tissue. Chua et al., (2021) have proposed a single-time frame convolutional neural network for the estimation of force using RGB images and robot states and did validation on dragon skin. However, the use of distal sensors has resulted in reduced accuracy in this type of network. A novel force estimation has been proposed (Hooshier et al., 2021), a magnetostriction-based modality based on frictional torque generation between a hollow cylindrical MRE and a ferromagnetic shaft for robotic interaction force application. But the authors tested only an artificial silicon model (EcoFlex 00-50) and did not consider any material or biological properties of the material that resulted in the failure of estimation of force in the various layers of the skin. A predictive modeling approach was proposed by (Batty et al., 2022) to model and predict haptic force feedback in RAMIS. This method utilizes the exponentially weighted recursive least squares (EWRLS) algorithm to determine the necessary parameters for force modeling. However, it is important to note that this study specifically focuses only on homogeneous phantom soft tissue. However this work is only rely on homogeneous soft tissue. Sabique et al. (2022) have proposed a stereovision-based force estimation based on stiffness mapping in surgical tool insertion using the LSTM-RNN network. However, the authors have not reported any changes in estimated force in various orientations and angles of insertion with respect to various tool diameters. Recently, (Ko et al., 2022) have presented a vision-based force estimation using an RGB depth image of an object and robot, robot position, and electrical current by combining DenseNet and Transformer encoder/decoder for robot grip motion (Wu et al., 2022) in the absence of any force or tactile sensor. The focus of the work was on hard and soft objects without consideration of any material properties. The method produced an error of 3 %, which is undesirable in robotic-assisted surgery. A summary of the research carried out in the area of learning and vision based methods is shown in Table 1.

Realistic surgical experience in soft tissue interaction is one of the most challenging parts of virtual surgical simulation and real-time surgery. All the works mentioned above, except a few, are based on discrete palpation. Most of the early addressed research did not meet expectations due to the estimation of the force components

based only on tissue deformation on surface properties (Angelica et al., 2015, Aviles et al., 2015a, Aviles et al., 2017, Aviles et al., 2016, Aviles et al., 2015b, Aviles and Casals, 2014., Aviles et al., 2014), (Marban et al., 2019), (Abeywardena et al., 2019) and (Chua et al., 2021), use of joint tension and kinematics (Edwards et al., 2021, Jung et al., 2021), and use of multiview and image depth (Ko et al., 2022) which fail in the event of occlusion once the tool had pierced the tissue. More specifically, significant biological features in the estimation of the force parameter during real-time surgery were not taken into account (Hooshier et al., 2021, Tatinati et al., 2015, Yu and Wang, 2015, Zhang et al., 2018). (Abeywardena et al., 2019, Aviles et al., 2017, Aviles et al., 2014, Chua et al., 2021, Edwards et al., 2021, Hooshier et al., 2021, Jung et al., 2021, Ko et al., 2022, Marban et al., 2019, Mendizabal et al., 2019) have omitted some of the relevant feature sets, such as angle of insertion, orientation except (Aviles et al., 2017), stiffness, and diameter of the tool, which have a significant role in real-time surgery. (Marban et al., 2019) found inadequacy in the quality of the evaluated force while using only tool or vision data. To the best of the researcher's knowledge, none of the authors has considered biological features like stiffness, insertion depth, and damping coefficient (predefined value) in training the neural network. Vision has the limitation of its inability to identify the parametric skin value based on deformation at any time following the incision of the needle or surgical tool incised due to occlusion. Due to the complexity of data extraction, very few works (Marban et al., 2019, Noohi et al., 2014) have reported force estimation task experiments on real tissue such as lamb and porcine; (Aviles et al., 2014, Giannarou et al., 2016, Kennedy and Desai, 2005, Kim et al., 2012) have done validation on the silicone-based artificial skin or organ. The overall conclusion from available literature is that the problem of estimation of the force parameter of tissues at various layers given tool penetration depth with different tool diameter, orientation still remains unanswered.

This work has proposed a real-time model for the estimation of the applied force at various layers of the *in-vivo* porcine skin, and the same data was logged onto the simulated environment for Visuo-haptic surgical feedback. Based on the training dataset, the proposed method is suitable for both the surfaces and the internal layers of the tissue, skin, organ, etc. The Saint-Venant-Kirchoff elastic model (Haouchine et al., 2018) was used for the determination of stiffness at various layers following the prediction of reaction force. A surgical device has been modeled for the demonstration of force seen during various surgical tasks and validation at a later stage. Building on previous contributions, this paper proposes a novel method for the reproduction of variable haptic force feedback and simulation across different layers of tissue. The following significant additions have been made:

- This work has collected real-time data from original tissue in different dimensions (angle, orientation, tool diameter, torque, rotation, insertion depth etc.) and biological features like (stiffness and damping coefficient), used in real-time and virtual environments in estimating the force.
- This work has proposed an LSTM-RNN + DR + CLR -based neural network framework for implementation in real-time and the prediction of force across various layers.
- The proposed method has the ability to determine the reaction force on the surface as well as in various layers of the tissue based on the training dataset.
- The calculated stiffness, derived from force measurements in relation to position, is saved as volumetric data within the virtual environment to enable a realistic surgical experience in the virtual environment.
- This study recommends dimensionality reduction and cyclic learning rate-based optimization to optimize the dataset and learning rate with fewer datasets and minimal computational cost. Additionally, an RMSE-based loss function has also been explored to create a smooth force data model.

The paper is structured as follows: Section 2 provides a detailed description of the methodology used. Sections 3 and 4, respectively cover the Force estimation module and Visuo-haptic module. The evaluation scheme and results are presented in Sections 5 and 6, respectively. The paper concludes with the future direction for research.

Section snippets

Methodology

The major objective of this work is to replicate the variable haptic force feedback in the surgical task in both visuo-haptic and real-time surgical procedures. The initial step is to collect the porcine and artificial tissue samples. The second step is to prepare samples suitable for conducting the experiment. The third step involves designing an appropriate experimental setup to conduct the experiment and perform validation. Validation of the result required the collection of significant

Force estimation module

The main content of the force evaluation module is the LSTM-Recurrent neural network algorithm, which helps the prediction of the variable force in tissue layers based on various feature sets. The following Eq. (1) was used for the calculation of force.

$$= +$$

Here K stands for stiffness matrix, usually determined from the elastic parameters of the material, such as Young's modulus (E) and the dimension of the material (Haouchine et al., 2018), x is the displacement or depth of the surgical

Biomechanical modeling of human skin

After reviewing the literature on skin modeling, it became evident that the human skin is one of the most challenging materials to model in computer graphics. This is due to the complexity of the skin material, which varies from individual to individual based on location, sex, color, texture, shape, and time (Jablonski, 2021). The skin is composed of three significant layers: the Epidermis, Dermis, and Hypodermis. The skin covers an area of 1.2–2.0 m² in adults and represents 12 %–16 % of the

Evaluation scheme

The following processes are related to evaluation schemes.

- Analysis of data collected from the porcine skin.
- Estimation of signal quality using PCC and RMSE metrics for six different cases are described in Table 4 for all three datasets.
- An assessment was done for the determination of the associated strength between the estimated force (output) in real-time and the Ground Truth Value (GTV) on test data for all three datasets.
- The predicted force was evaluated and compared with the ground truth

Analysis of dataset collected from the porcine skin

Fig. 8 depicts the parametric plot of three layers of skin: stratum corneum, epidermis, and dermis, and complete profiling of the porcine skin layer in both insertion and retraction tasks. The above experiment provided the key metrics for the determination of the mechanical behavior of the porcine skin, which included Stiffness (S), the Force of insertion (F_i) to break the stratum corneum, related to Ultimate Tensile Stresses (UTS), the Depth of insertion (D_i) to rupture the SC and Work done (W)

Conclusion and future work

This paper deals with a detailed study made on the online estimation method for predicting variable interaction forces in real and virtual environments. Machine learning-based techniques have been proposed (RNN-LSTM + DR + CLR) for the estimation of the force in a real-time environment. Considering the requirement of only the training data, the suggested method is appropriate for various medical procedures. In the initial stage, the pulling and pushing procedures were tested. The surgical

Compliance with ethical standards

Conflict of interest: This study was funded by the University Grants Commission (F./2017-18/NFO-2017-18-OBC-KER-60500), The Government of India.

Ethical approval: All applicable international, national, and Anna University guidelines for the care and use of animals were followed, and all the experiments were directed on the authority of strategies by the Animal Welfare Board of India.

Informed consent: Informed consent was obtained from all individual participants included in the study. For this

CRedit authorship contribution statement

P.V. Sabique: Conceptualization, Methodology, Data curation, Software, Formal analysis, Investigation, Visualization, Writing – original draft. **Ganesh Pasupathy:** Supervision, Conceptualization, Writing – review & editing. **Sivaramakrishnan Ramachandran:** Supervision, Conceptualization, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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